

E-commerce Performance Review

Sales, delivery reliability and customer experience

January 1–December 31, 2025 | 15,000 orders | 7,807 customers

Executive conclusion

Delivery reliability is the clearest business problem. 59.58% of non-cancelled orders arrive late and 13.20% are severely delayed. Severe delays coincide with a sharp rating decline and more returns. The issue is broad-based across carriers, cities and product categories, rather than concentrated in one obvious segment.

Core KPI	Verified result
Recorded net order value	35.27 million CU
Retained order value proxy	32.29 million CU
Late orders / non-cancelled orders	8,695 / 14,593 = 59.58%
Severely delayed / non-cancelled	1,927 / 14,593 = 13.20%
Returns / all orders	851 / 15,000 = 5.67%
Cancellations / all orders	407 / 15,000 = 2.71%
Average rating / net average order value	4.02 / 5 2,351.22 CU

Priority actions: instrument delivery milestones, investigate severe-delay causes, and reconcile cancelled/returned orders with payments and refunds. Do not replace carriers solely on these observed late rates.

Interpretation: Currency is not supplied. CU means currency units. Recorded net value is order value after discounts, not recognized revenue. The retained proxy is not cash collected or profit.

1 | Data integrity and metric definitions

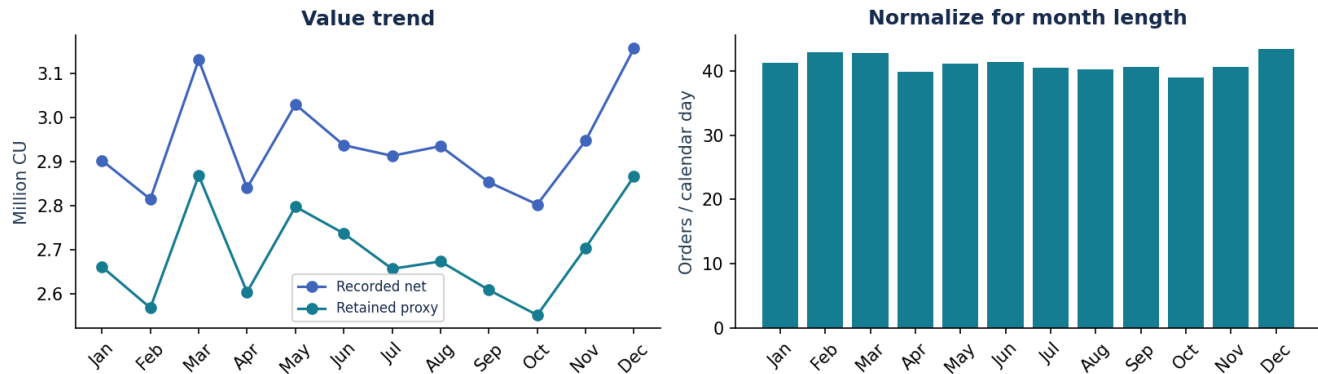
The dataset covers all 365 order dates in 2025, across 10 cities, 8 product categories, 5 warehouses and 5 shipping partners. All 15,000 order IDs are unique; all 17 source columns have zero missing cells.

Check	Result / implication
Net value calculation	Matches $\text{OrderValue} \times (1 - \text{DiscountPct})$, within 0.005 CU rounding tolerance.
Delivery labels	Non-cancelled on-time orders have zero delay; Delayed = 1–2 days late; Severely Delayed = 3–11 days late.
Cancellation consistency	All 407 Cancelled statuses match the cancellation flag.
Overlapping outcomes	29 orders are both returned and cancelled. Event order is not available; investigate rather than assume invalid.
Cancelled delivery fields	Cancelled records still have delivery durations and ratings. Exclude them from delivery reliability denominators.
Potential data-generation concern	Balanced segment volumes and nearly identical carrier rates merit validating provenance and event definitions; synthetic origin cannot be determined from this file.

Definitions: Late rate uses $\text{ActualDeliveryDays} > \text{PromisedDays}$ among non-cancelled orders. Return and cancellation rates use all orders unless explicitly stated. Retained value excludes the union of cancelled and returned flags; 29 overlapping records are removed once.

Limitations: No currency label, costs, margin, refund amount, payment status, shipment timestamps, delivery date, acquisition spend, inventory or SLA revision history. Ratings have no missing values even for cancellations, and their collection process is unknown. No causal, profitability or marketing ROI conclusions can be drawn.

2 | Commercial performance and trends



Recorded gross order value: 42.81M CU. Discounts total 7.54M CU, or 17.61% of gross value (value-weighted). Recorded net value is 35.27M CU; net average order value is 2,351.22 CU.

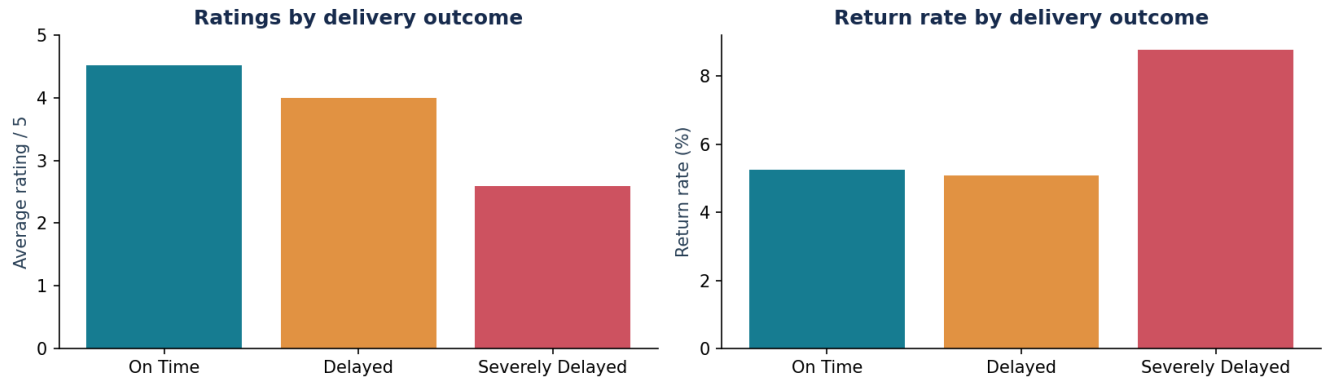
Stable overall volume, not convincing seasonality: Monthly recorded net value ranges from 2.80M CU in October to 3.16M CU in December. December is 8.79% above January, but Q4 is only 0.67% above Q1. Daily order volume ranges from 38.94 in October to 43.35 in December. February has fewer total orders partly because it has only 28 days.

Category mix: Fashion contributes 7.03M CU (19.93% of recorded net), followed by Electronics at 6.01M CU (17.05%) and Home & Kitchen at 5.45M CU (15.47%). Combined, these categories contribute 52.44%. Their volume makes them useful operational pilot segments, but these data do not establish stronger margins or incremental demand.

Customer repeat purchasing: 4,355 of 7,807 observed customers (55.78%) place at least two orders within the year. This is within-window repeat purchase incidence, not cohort retention, loyalty uplift or customer lifetime value.

Promotion interpretation: Discount percentage has no clear monotonic relationship with returns among non-cancelled orders (Spearman rho = -0.0082, p = 0.321). The file cannot establish discount elasticity because it has no randomized offers, visits or conversion data.

3 | Delivery reliability and customer experience



Outcome	Orders	Rating	Return rate
On Time	5,898	4.52	5.24%
Delayed (1–2 days)	6,768	4.00	5.08%
Severely Delayed (3+ days)	1,927	2.60	8.77%

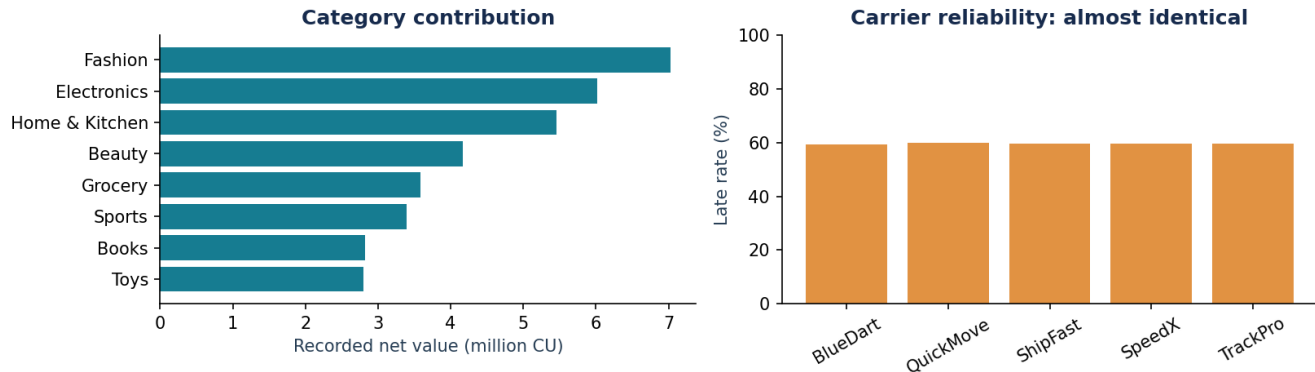
Severe delays matter most: Severe-delay return rate is 3.53 percentage points higher than on-time (approximate 95% confidence interval: +2.15 to +4.92 points), or 1.67 times as high. Across non-cancelled delivery outcomes, return-rate differences are statistically detectable (chi-square $p = 1.10 \times 10^{-9}$).

Ratings: On-time orders average 4.52 compared with 3.69 for all late orders and 2.60 for severely delayed orders. Severe vs on-time is a 1.93-point difference. Mildly delayed orders do not have a higher return rate than on-time orders in this sample, so avoid treating all delays as equally harmful.

Reliability target: On-time rate is only 40.42%. A reduction in late rate from 59.58% to 50% at the same annual volume would mean approximately 1,399 fewer late orders. This is a target scenario, not a forecast. Monitor absolute delivery time too, to avoid improving the metric only by extending promises.

Causality caveat: Product issues, customer expectations or other unobserved factors may affect both returns and ratings. Use controlled operational pilots to estimate the actual effect of faster delivery.

4 | Segment insights and decision guardrails



No clear winning carrier: Non-cancelled late rates range from 59.42% (BlueDart) to 59.77% (QuickMove), a spread of only 0.35 percentage points. A chi-square test detects no carrier association with lateness ($p = 0.999$; Cramer's $V = 0.0024$). Do not infer that the brand names represent external carrier performance; these results apply only to this dataset.

Warehouses: WH-BLR has the highest observed late rate (60.95%) and WH-HYD the lowest (57.88%). The overall warehouse association is not statistically detectable at the 5% level ($p = 0.095$). Investigate process differences, but do not label warehouse performance as established on these rankings alone.

Geography and categories: Late rates are also broadly similar across cities (chi-square $p = 0.681$) and categories ($p = 0.966$). City late rates span 57.77%–61.42%; category late rates span 58.67%–60.44%. These exploratory comparisons are not multiplicity-adjusted.

Distance does not explain delay in this file: Spearman association between DistanceKm and days late is essentially zero ($\rho = 0.00007$, $p = 0.994$). There is no evidence here to prioritize long-distance lanes as the main bottleneck; inspect actual pickup, dispatch and last-mile timestamps before redesigning the network.

Descriptive watchlist: TrackPro has the highest all-order return rate (6.48%) and cancellation rate (3.31%). Warangal has the highest city return rate (6.95%). These are monitoring signals, not causal attribution; compare case mix, failure reasons and uncertainty before intervention.

5 | Value exposure and testable opportunities

Value bridge	Million CU
Gross order value	42.807
Less discounts	7.539
Recorded net order value	35.268
Less cancelled order value	0.953
Less returned value among non-cancelled orders	2.022
Retained order value proxy	32.293

Flag-associated value exposure: 1,229 unique orders are returned, cancelled or both. Their combined recorded net value is 2.975M CU (8.44% of total recorded net). This is not a verified loss amount: partial refunds, recovery, reorders, cancellations before payment and return resale are unknown.

Illustrative severe-delay scenario: If severely delayed orders had the same return rate as on-time orders, there would be about 68 fewer return flags. At the severe-delay segment's average net order value of 2,347 CU, that corresponds to about 160,000 CU of illustrative order-value exposure. This applies an observational difference as a hypothetical assumption; it is neither causal savings nor a forecast.

Illustrative discount scenario: Reducing discount percentage by one absolute point across unchanged gross order value would add approximately 428,070 CU to recorded net order value mechanically. Demand, order mix and returns could change; measure conversion and contribution margin in a randomized promotion test before implementing.

Business data needed next: Payment/refund ledgers for realized value; COGS and shipping costs for contribution margin; delivery milestones and failure reasons for root cause; campaign exposure and site conversion for promotion effectiveness; inventory and stockouts for capacity planning.

6 | Prioritized action plan and measurement

When / owner	Action	Measure
First 2 weeks / Data + Finance	Reconcile cancelled and returned records, audit 29 overlaps, and document currency and event definitions.	Payment/refund reconciliation rate; overlap reasons; metric sign-off.
Weeks 1–4 / Operations	Add order-to-dispatch, pickup, hub, out-for-delivery and delivery milestones. Review severe-delay cases across all carriers.	Severe-delay rate, 90th percentile lateness, scan completeness.
Weeks 3–8 / Operations + CX	Run a controlled severe-delay prevention pilot in high-volume categories; test exception alerts and early customer outreach.	Late rate, severe-delay rate, returns and ratings; compare against a control.
Weeks 4–8 / Commercial + Finance	Test discount changes only after adding conversion and unit economics.	Conversion, net value per visitor and contribution margin, not AOV alone.
Ongoing / Analytics	Review monthly daily-normalized demand and segmented delivery reliability with sample sizes.	Recorded vs retained value; non-cancelled late rate; return/cancel rates.

Dashboard guide: Filter by month, city, category, warehouse and shipping partner. Every chart and KPI updates to the selected slice. The scorecard can be grouped by city, category, warehouse or carrier. Filtered small samples can fluctuate substantially; dashboard comparisons do not include significance tests.

Monitoring guardrails: Keep cancelled orders out of delivery denominators, distinguish severe from mild delays, report orders alongside rates, and track promise changes. Alert thresholds should be validated prospectively rather than inferred from one retrospective year.

Analysis method: Descriptive aggregates; chi-square contingency tests for categorical differences; Spearman rank correlation for monotonic associations; independent binomial normal approximation for the severe-vs-on-time return-rate difference. Tests are exploratory and unadjusted for multiple comparisons; the file may not be a random sample of future operations.

Bottom line: Fix measurement and severe-delay processes before reallocating carriers, claiming seasonal growth, or optimizing discounts without margin data.